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# Co-Design at the Boundary: Understanding the Dynamics of Open Innovation Between Companies and Communities

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## ABSTRACT

For over two decades, firms have built platforms and engaged open innovation communities to improve and customize their products through widened participation in the design process. While the benefits of involving those outside a firm as co-designers have been well described, how co-design processes unfold over time at the boundary between those in an organization and an open online community remains poorly understood. Using the theoretical lens of boundary work and a five-year ethnographic study with a firm that pioneered co-design in the automotive industry, we develop a theoretical model that shows how and why co-design between the firm and its open online community evolved over time. We describe three different types of co-design (intimate, instrumental, and limited co-design) and show how these are intertwined with existing types of boundary work (collaborative, configurational, and competitive boundary work). Importantly, we unravel the dynamic shifts between different types of co-design and associated boundary work, showing the driving forces that precede shifts. Our research extends the boundary work literature by proposing forces that prompt organizations to shift between types of boundary work, illustrating how these forces are tied to characteristics of the environment and open community and are integral to the work of the organization. More broadly, this study contributes new insight into the temporal evolution of design collaboration between organizations and online communities.

## 1 | Introduction

Over the past two decades, organizations ranging from large firms like BMW, LEGO, and NASA to smaller firms like Threadless and Quirky have pursued open innovation by developing open online communities of customers, enthusiasts, and experts to help design new products and services (e.g., Brabham 2010; Terwiesch and Xu 2008; Lifshitz-Assaf 2018; Riedl and Seidel 2018; Hofstetter et al. 2021). Many of these examples involve co-design, that is, external partners acting as direct co-producers of designs, through an online platform directly managed or controlled by the firm (e.g., Langner and Seidel 2015; Riedl and Seidel 2018). Despite significant investment from major firms, not all co-design efforts with online communities have led to commercial breakthroughs, resulting in many such efforts being abandoned (Pedersen et al. 2022). A possible explanation lies in the unpredictability and fluctuation in the amount, frequency, and quality of the contributions coming from external contributors (West and Bogers 2014) and in the lack of preparedness of companies to organize, implement, and sustain collaboration with open online communities over time (Lichtenthaler et al. 2011; Amann et al. 2022). As firms and their open platforms struggle to maintain a committed community of external contributors, how to effectively sustain co-design with

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**Managerial Summary**

Investing in boundary work is important for sustaining open innovation over time, and the type of boundary work performed influences the dynamics and outcomes of co-design. Boundary work can take multiple forms–whether collaborative, configurational, or competitive—and a firm can tailor its boundary work based on its innovation strategy, prevailing market forces, and the readiness of its open innovation community at any given moment. For example, if a firm employs an innovation strategy focused on exploiting established ideas, it might pursue configurational boundary work involving relatively infrequent, formal, and structured feedback alongside strategic communications with community members around tightly scoped open innovation challenges. Alternatively, if employing an exploration focused innovation strategy, a firm might engage in collaborative boundary work, providing frequent, informal, personalized, and relationship-oriented feedback to community members around relatively open-ended innovation challenges. Additionally, it is important for firms to monitor the evolution of their community over time. Changes in community composition, competencies, and expectations or in a firm's needs can shift the readiness of a community and firm to effectively engage in co-design. Overall, there is no single best approach, but being intentional about boundary work can help firms and communities to more fruitfully design together.

communities and manage the boundary between the community and company over time, remains an open question.

The present study focuses on the evolution of co-design efforts at the boundary between a firm and an open online community. Research underscores the importance of managing the boundary between an organization and an external community in open innovation collaborations (e.g., Wilhelm and Dolfsma 2018; Randhawa et al. 2016; Ollila and Elmquist 2011) and highlights impediments to successful open innovation efforts, such as inflexible organizational cultures and professional identities or inadequate governance models and project management processes (West and Bogers 2014; Saura et al. 2023; Lifshitz-Assaf 2018; Ahlfänger et al. 2022; Cricelli et al. 2023). Researchers have started to investigate how collaborations between firms and communities evolve over time (e.g., Fuge and Agogino 2014; Liu et al. 2022; Lindberg et al. 2022; Pinarello et al. 2022). However, these studies tend to take the perspective of either the company or the community, not both. Research has yet to characterize how the boundary between an organization and a community is co-constructed and influenced *over time*, that is, how organizations and communities engage dynamically in "boundary work" (Langley et al. 2019), leaving us with a limited understanding of how to make open innovation initiatives more sustainable. To advance knowledge on these issues, and to make sense of what happens at the boundary between a firm and an online community, we draw upon the theoretical perspective of boundary work. Organizations, groups, and individuals perform boundary work when they work across boundaries to influence other groups, occupations, and organizations with which they are intertwined. Boundary work defines how people are included or excluded in the dynamics of collaboration over time (Langley et al. 2019; Lamont and Molnár 2002; Phillips and Lawrence 2012), which is ultimately tied to cross-boundary learning and performance (Lindberg et al. 2017; Mørk et al. 2012; Yagi and Kleinberg 2011; Zietsma and Lawrence 2010).

While the theoretical perspective of boundary work appears relevant to the dynamics of co-design and open innovation, few studies have applied boundary work to help explain such dynamics. Using the lens of boundary work (e.g., Kellogg et al. 2006; Langley et al. 2019; O'Mahony and Bechky 2008), our study aims to answer the following motivating question: *How and why do co-design practices between organizations and online communities sustain or diminish over time?*

To address this question, we conducted a five-year ethnographic study (2014–2019) with Community Cars (a pseudonym that we shorten to Com-Car), a firm that pioneered co-design in the automotive industry by building an online platform and hosting an open online co-creation community. Com-Car is unique in the high degree to which co-design was part of its core strategy. We investigated the dynamics of co-design at the boundary between Com-Car and its online community using a grounded theory approach and Com-Car as an extreme, and thus illuminating, case (Siggelkow 2007).

Taken together, our theoretical model shows co-design and boundary work dynamics over time between Com-Car and its online community. We describe three types of co-design–intimate, instrumental, and limited co-design—and show how these co-design types occur alongside three established types of boundary work. We also identify forces that appear to drive shifts between the three co-design and associated boundary work types. In doing so, we aim to understand the dynamics of collaboration between the company and its open online community but not assess the merits or outcomes of the boundary work or co-design forms. We follow the journey of Com-Car as they begin with a strong commitment to co-design with their co-creation community and shift over time between different forms of co-design and boundary work before ultimately letting go of their online community and closing their co-design platform. Our theoretical model provides some explanation for these dynamics, contributes to the co-design literature by offering a typology of co-design that is tied to established categories of boundary work, and extends the boundary work literature to show shifts between boundary work types and the forces that may drive such dynamics. We close with managerial implications for innovation practitioners and open innovation communities and the sociotechnical infrastructures that support them.

## 2 | Theoretical Framework

To examine how co-design is sustained or diminished over time, we draw upon literature on co-design between companies and online communities alongside literature on boundary work.

### 2.1 | Co-Design Between Companies and Online Communities

The literature on co-design between firms and online communities draws from two often parallel bodies of work: co-design



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within the engineering and design fields and open innovation within the innovation management field. Since both are relevant to our study, we will briefly introduce each. From a design perspective, co-design between an online community and a firm differs from other forms of open innovation in that external contributors are engaged not just as sources of ideas or feedback but as iterative co-developers of designs (e.g., Ramaswamy and Ozcan 2014). Instead of a consultative, passive, or extractive design process, co-design involves a co-constructive and participatory design process that encourages contributors from an external community to make direct design contributions and decisions (Arnstein 1969; Mazzurco et al. 2018; Sanders and Stappers 2008).

In the design literatures, co-design is often associated with participatory design, a literature that has a history of both studying and building tools for engaging stakeholders in the collaborative design process (e.g., Kensing and Blomberg 1998; Schuler and Namioka 1993; Simonsen and Robertson 2012; Wacnik et al. 2025). Scholars have examined the quality of co-design dialogue between designers and potential users or members of an online community. These studies have illustrated the difficulty of generalizing user needs based on what users say, ways of making tacit knowledge explicit, the importance of constructive conflict, and other conversational competencies of designers (Buur and Larsen 2010; Luck 2007). More recent studies have attempted to unpack the notion of participation in co-design, including identifying guiding principles for co-design between firms and online communities, such as mutual learning, democratic processes, and equalizing power relations between firms and community members (Andersen et al. 2015; Bratteteig and Wagner 2016; Luck 2018).

In the innovation management literature, co-design has been found to have potential benefits at both the individual and organizational levels, though these benefits are not always realized. At the individual level, co-design can lead to mutual cross-boundary learning (Lundmark 2018) and professional identity transformation, for example, transforming NASA R&D professionals from "problem solvers" to "solution seekers" (Lifshitz-Assaf 2018). At the firm level, co-design can improve product adoption (Gemser and Perks 2015), firm reputation (Fuchs and Schreier 2011), and product variety (Al-Zu'bi and Tsinopoulos 2012). In some studies, co-design with external contributors was positively associated with innovation performance (e.g., Faems et al. 2010; Ferreras-Méndez et al. 2015; Tsai 2009), while in others it was negatively associated (e.g., Knudsen 2007; Knudsen and Mortensen 2011). These mixed results may stem, in part, from co-design performance being influenced by which external contributors are engaged, when they are engaged, and to what degree. In a survey of 126 co-design projects, Mahr et al. (2014) found that co-design was most successful when developing products that were highly relevant to contributors' needs and experiences and that co-design with lead users yielded both novel and relevant product concepts. Overall, involving external contributors in design and development has been shown to have an "inverted U" relationship with innovation performance, with both too low or too high levels of involvement being tied to poor innovation performance (Belderbos et al. 2010; Laursen and Salter 2006).

Most studies on the relationships between those involved in co-design across firms and online communities have focused on cross-sectional data, examining specific contexts or challenges in a short period of time. However, a few have begun to explore the role of time in these collaborations. Temporal research in this area typically falls into two areas: community evolution or organizational practices in community management over time. Studies in the first area examine how community social networks and structures evolve over time (e.g., Fuge and Agogino 2014). Some note that the successes of community members in prior challenges influence their continued participation in future ones (Liu et al. 2022; Lindberg et al. 2022). The second area of research focuses on the mechanisms that organizations establish to sustain collaboration over time (e.g., Antikainen et al. 2010). Pinarello et al. (2022) highlight that the adoption and use of open innovation communities within firms is not linear, being influenced by multiple factors, such as "(i) the opportunity to mitigate innovation risks, (ii) the opportunity to leverage external, more unfamiliar competencies, and (iii) the option of relying on external facilities or tools" (p. 558). Maintaining contributions from an online community can be challenging and often requires specific efforts, such as appointing community managers to act as intermediaries (Randhawa et al. 2017).

Taken together, the existing literature tends to consider either the perspective of the organization or the community members without focusing on the boundary between them. To fully understand collaboration between a firm and an online community—not only how ideas are generated but also which ones are implemented or discarded over time (West and Bogers 2014)—we need to examine what occurs at the firm-community boundary over time. We draw upon boundary work theory to gain deeper insights into how organizations and communities co-design together through time.

## 2.2 | Boundary Work

Boundary work offers a theoretical framework that can help to understand why and how co-design between those internal and external to an organization shifts and evolves over time. Boundary work is defined as intentional communication or action to influence "the social, symbolic, material, or temporal distinctions affecting groups, occupations, and organizations" (Langley et al. 2019, 704; Lamont and Molnár 2002; Phillips and Lawrence 2012). In her study of nurses and doctors' daily work in a hospital unit, for instance, Liberati (2017) shows that nurses dismantled professional boundaries between themselves and doctors by joining doctors' conversations and autonomously establishing their own duties, actively sharing information about patients, and participating in the diagnosis process.

Boundary work matters because it influences who and how people are included or excluded in collaborative efforts and the dynamics of collaboration over time. Boundary work can perpetuate or disrupt power relations between groups, professions, etc. (Arndt and Bigelow 2005; Barrett et al. 2012; Bucher et al. 2016) and, ultimately, influence cross-boundary learning and performance (Lindberg et al. 2017; Mørk et al. 2012; Yagi and Kleinberg 2011; Zietsma and Lawrence 2010).



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While the theoretical perspective of boundary work appears relevant to the dynamics of co-design and open innovation, few studies appear to have applied boundary work to help explain dynamics in co-design and open innovation practice. Most of the studies on boundary work have focused on the interactions between occupations (e.g., professional groups in healthcare, Liberati 2017), groups within and between organizations (e.g., firms in interorganizational supply chains), and large social groups (e.g., in scientific inquiry, Garud et al. 2014).

Such studies have been collected by Langley et al. (2019) in an integrative review of the boundary work literature. They identified three overarching categories: competitive, collaborative, and configurational boundary work. Competitive boundary work has roots in Bourdieu's "fields of practice" (1977) and involves the construction or defense of group, occupational, or organizational boundaries as a means of acquiring or reproducing power, position, status, and resources. It involves activities like defending, contesting, and creating boundaries. For example, Gieryn, who originally coined the term "boundary work" (1983), studied the rhetorical moves used by scientists (i.e., emphasizing rigor, objectivity, and reliance on causal principles) to demarcate science from nonscience and thus afford scientists, as a bounded occupational group, positions of authority and credibility over other fields (1983).

Collaborative boundary work pertains to the activities in which different groups, for example employees in a firm and designers in an online community, engage at their interface to establish and maintain collaborative and coordinated patterns in situations where collective goals cannot be attained by any single group alone (Langley et al. 2019; Kellogg et al. 2006). Collaborative boundary work is based on the negotiated order perspective which posits that social order, such as a division of labor, is developed through ongoing negotiations between those involved in the work (Strauss 1978; Strauss et al. 1963). In competitive boundary work, boundaries are seen as barriers. In collaborative boundary work, boundaries are more like junctures to be drawn upon to facilitate, rather than inhibit, collaboration (Quick and Feldman 2014), leading to improved mutual understandings about who will do what and when. Collaborative boundary work involves activities like negotiating, aligning, embodying, and downplaying boundaries to get cross-boundary work done (Apesoa-Varano 2013; Barrett et al. 2012; Bechky 2006; Quick and Feldman 2014). Prior work on open innovation touches on aspects of collaborative boundary work when it describes the challenge and importance of collaboration between parties. For example, studies by Barbic et al. (2021), Wilhelm and Dolfsma (2018), Ollila and Elmquist (2011) underscore how knowledge sharing between a firm and an open online community is crucial, but fragile, in complex open innovation projects.

Configurational boundary work, a third form of boundary work, introduced by Langley et al. (2019) describes actions in which managers, entrepreneurs, and other leaders "work to reshape the boundary landscape of others," (p. 720), guiding the development of both competitive and collaborative boundary work patterns. In essence, configurational boundary work involves the strategic use of boundaries to alter or reorganize interaction patterns, manipulating the integration of various groups to bring specific activities together while temporarily keeping others apart to facilitate effective collective action. This involves arranging, buffering, and coalescing boundaries (Langley et al. 2019). Empirical evidence from open innovation contexts suggests the presence of configurational boundary work when describing a need to manage trade-offs between open and closed innovation over time (Barbic et al. 2021; Ahlfänger et al. 2022) or between creating versus capturing value (West and Bogers 2014; Zobel and Hagedoorn 2020). In these situations, gatekeepers and boundary organizations—external actors not directly involved with the work happening at the boundary—influence the pattern of separation and integration of open innovation over time (e.g., Randhawa et al. 2016; O'Mahony and Bechky 2008; Wilhelm and Dolfsma 2018).

In their review, Langley et al. (2019) suggest foci for future boundary work research including examining its temporal dynamics, shifts between its forms, interconnectedness of boundaries, and boundary work in new boundaryless kinds of organizations. Prior work illustrates shifts from configurational to competitive boundary work (Granqvist and Laurila 2011; Grodal 2018; Liao 2016) and from competitive to collaborative and then to configurational boundary work (Mikes 2011; Quick and Feldman 2014; Zietsma and Lawrence 2010). While such sequences of boundary work have been observed, what drives these shifts and temporal dynamics between boundary work forms remains unclear.

The present study aims to advance the state of knowledge around the dynamics of boundary work between organizations and online communities. We do so by analyzing the evolution of co-design and associated boundary work which yields a typology of co-design that connects to established forms of boundary work. We also present a theoretical model that helps to explain the shifts between forms of co-design and associated boundary work, including forces that appear to drive shifts between forms.

## 3 | Method

We collected ethnographic data over five years (2014–2019) at Community Cars (a pseudonym, hereafter "Com-Car")—a firm that pioneered co-design in the automotive industry—and during and after this period developed a theoretical model using a grounded theory approach. The following sections describe our research site, data collection, and analysis methodology.

### 3.1 | Field Site: Com-Car

We collected ethnographic data at Com-Car (pseudonym), a United States-based automotive start-up founded in 2007 on the promise of co-creating innovative boutique motor vehicles. Com-Car pioneered co-design in the automotive industry by being among the first automotive firms to build an open innovation community. Com-Car's founder and CEO started the community with the strategic objective of facilitating co-creation between online crowds and internal employees, including designers, engineers, marketing, and IT specialists. Initially, as we



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will show in the findings, the boundary between Com-Car and its open online community was established through quite intentional collaborative boundary work.

The members of Com-Car's open online community were designers, engineers, students, and car enthusiasts. At its peak, the Com-Car community consisted of over 80,000 members globally. Some were motivated to join by the prospect of contributing to automobile designs that might actually go into production. Others found value in the feedback their work received from other community members and Com-Car employees or in the visibility offered, leading to job opportunities in companies like BMW, Peterbilt, and Com-Car itself. Others, including car enthusiasts and students, sought to enhance their skills in vehicle design. A few were exclusively drawn to the potential for winning prizes and awards. The least active participants were driven by curiosity and an interest in observing others' work. In terms of ongoing involvement, according to an internal Com-Car source, around 5% of their community members consistently participated in design challenges.

Com-Car's initial offerings were co-developed through close and frequent interactions with the community from the design stage through manufacturing. Com-Car regularly posted design challenges to the community, who responded with submissions of design ideas. These ideas were then workshopped by the community member(s) and Com-Car employees and eventually assessed by a panel of judges. Selected designs were absorbed into the Com-Car product development portfolio and further co-designed by community members and Com-Car engineers and designers across subsequent product development stages.

This extreme level of co-design appears unusual in hardware product development, especially as the core product development strategy of a firm. As an extreme case, Com-Car was an ideal setting for our inductive examination as it made underlying processes and mechanisms more "transparently observable" during grounded theory building (Pettigrew 1990, 275). Over time, co-design between Com-Car and members of their open community dwindled, and in 2020, Com-Car fully abandoned its open innovation community and co-design platform. To capture real-time decision-making and shifts in the co-design process, we triangulated data from longitudinal ethnographic observations, online posts, and interviews conducted at multiple stages over five-year period.

## 3.2 | Data Collection: Ethnographic Methods

Ethnographic methods are particularly appropriate to investigate processes and temporality in organizations (Langley 1999; Langley 2009). We conducted ethnographic observations and interviews (Spradley 2016; Lofland et al. 2022) at Com-Car over a five-year period (2014–2019). Our research team traveled to Com-Car's offices in the United States thirteen times, ranging from targeted one-day visits to a three-month stay, with most trips lasting for 3–5 days with roughly 8 h of observation per day. Visits to Com-Car were done solo and in pairs. The data were collected in two waves with monthly trips between June 2015 to March 2016, renewed trips in March 2019, and a three-month stay from June to September 2019. Our initial strategy was to select and follow multiple design challenges from the opportunity identification stage through product launch. However, this strategy proved infeasible for multiple reasons. First, the scope and duration of design challenges were relatively varied and unpredictable. Some challenges were for a whole car concept whereas others were for single parts or subsystems. Co-design on a challenge could last for days, weeks, months, or years with some challenges being extended while others being abandoned midway through. Finally, some challenges were related to products that were deemed strategic for the organization's development and growth, while others were considered minor experiments.

Thus, we decided to focus our main data collection on challenges, products, and projects that were central to Com-Car's strategy, employees, and community. During preliminary conversations with executives in 2014 and 2015 and subsequent interviews, we decided to focus on five vehicles as pictured on the timeline in Figure 1. We planned regularly spaced visits at times when important occurrences were happening in relation to these products. We adopted a strategy of observing any occurrences of design activity at Com-Car during our visits and paid special attention to interactions between Com-Car employees and community members. We also collected data through interviews during and in between our visits, and from the online co-design platform.

### 3.2.1 | Observations

Over the five-year period, our data include 80 days of in-person observations and field notes on-site at Com-Car offices and the main Com-Car "microfactory." When available, researchers were given a desk in Com-Car's open office layout. Observations occurred in the open office by researchers moving between different conversations, often in the context of displayed CAD models, design sketches, prototypes, and the like. Researchers also followed Com-Car employees into the nearby microfactory where substantial prototyping, problem solving, and low-volume production occurred. Regular all-hands and design team meetings were observed. Interactions between Com-Car employees and open community members occurred asynchronously via Com-Car's online platform and messaging platforms and, occasionally, synchronously through video conference. During observations, researchers asked questions to clarify issues and hear informants' interpretations of situations. We had lunch with informants and joined informal gatherings inside and outside of the office. Close attention was paid to the interactions and attitudes of Com-Car employees as they communicated with and about community members and each other. Detailed field notes were written on site at the end of each day. In some instances, meetings and informal interactions were audio recorded and later transcribed, and pictures were taken of prototypes, sketches, and manufactured vehicles.

We observed the activity of Com-Car designers and engineers whose tenure at Com-Car ranged from 6 months to 8 years and whose backgrounds ranged from mechanical engineering to industrial design and other traditional automotive fields. Com-Car executives and marketing professionals were also observed but to a lesser extent. With a few exceptions, all Com-Car staff



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presented as white American or European men between 25 and 60 years old, of whom several were hired from Com-Car's community.

### 3.2.2 | Interviews

We conducted 48 semi-structured interviews in English, lasting from 25 to 93 min. Thirty-nine interviews were conducted with 24 Com-Car employees, and nine interviews with members of Com-Car's open online community. Interviews were conducted both onsite and online. Following an ethnographic approach (Spradley 2016), interviews were open-ended with questions aimed at learning about informants' experiences with and perceptions of Com-Car design projects, working across the organizational boundary between Com-Car and community members, and the nature of their interpersonal relationships with those they engaged. Researchers solicited detailed explanations of project work including vehicle design challenges and solutions and how informants engaged in design work. Prompts for Com-Car employees included questions like: "Last time we visited, you were working on ___. How did that go?" and "How, if at all, are you engaged with community members right now?"

The nine community members who were interviewed encompassed Com-Car challenge winners, "old-timers" and "newbies," as described by our informants, and other contributors recommended by Com-Car employees as representing a wide set of perspectives. They included both individuals who were satisfied with their experience as community members as well as those who expressed discontent with interactions with Com-Car employees and co-design on the platform. The community members we interviewed were all men under 44 years old with training in fine arts, industrial design, engineering, theater design, or no design training (i.e., a high school student and a priest). Since community informants were globally distributed (e.g., in Italy, Romania, Saudi Arabia, and the United States), interviews were conducted through video calls. Our semi-structured interview protocol (Gubrium and Holstein 2001) included questions such as "Can you please describe the last challenge or activity you contributed to?," "Can you describe your interaction with other community members? ... with Com-Car employees?," and "Have you experienced any challenges in participating in the community? Could you give us an example?" Across all interviews, extensive follow-up questions were asked, and interview summary notes were written by researchers following interviews. Each interview was audio recorded and later transcribed. When possible, researchers collected diagrams, pictures, and other artifacts that informants offered as supplements. All informants and vehicles were given pseudonyms.

### 3.2.3 | Posts From the Com-Car Online Platform

Com-Car built an online platform that hosted both Com-Car-initiated design challenges, related exchanges of feedback, and more general automotive design forums. We collected an archive of written post content and artifacts (e.g., sketches, 3D and 2D designs) from Com-Car's online platform as evidence of co-design interactions between Com-Car employees and community members. We focused our attention on posts associated with design challenges and vehicle development programs that were also observed during our ethnographic observations and interviews at Com-Car (Figure 1). The number of community members who participated in this subset of challenges was 1278. For each of the observed challenges, researchers downloaded the design challenge brief, timeline, discussion forum, and design solutions submitted by community members (the "entries"). Each entry had a space for discussion where community members and the Com-Car team could post their comments and suggestions, thus supporting co-design. We analyzed entry discussions as additional data on co-design activities.

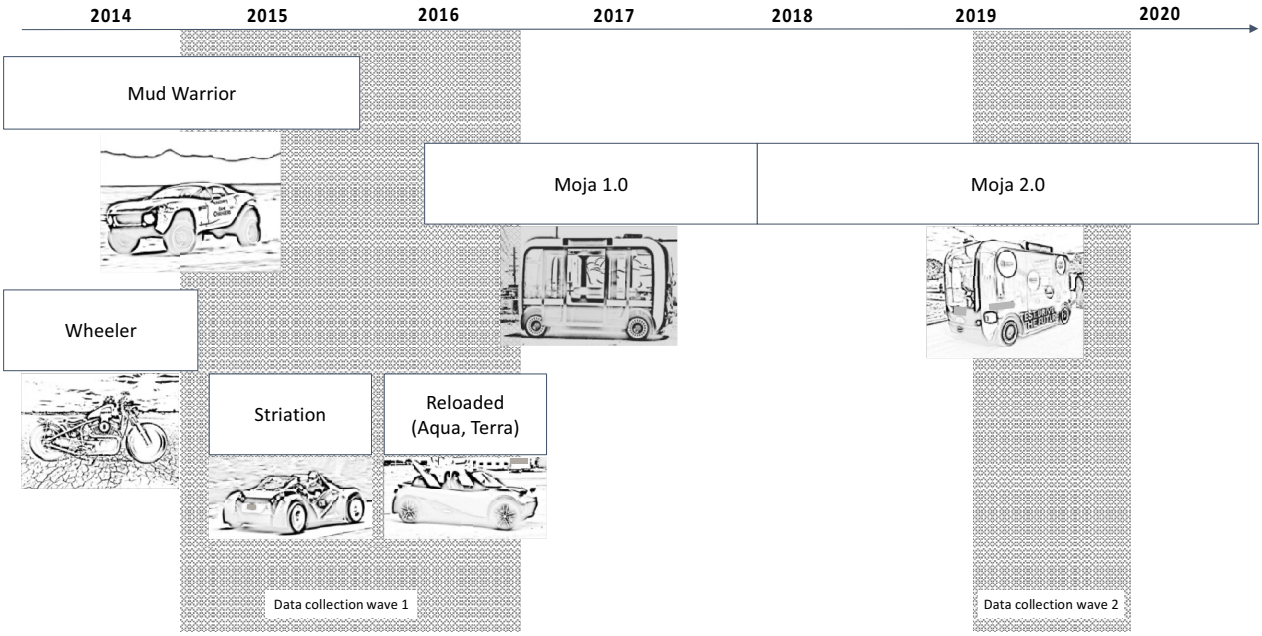


**FIGURE 1** | Select Com-Car product development programs observed. Hashed sections indicate the two primary waves of data collection. Images modified from Com-Car.



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### 3.3 | Data Analysis

Following a grounded theory approach (Glaser and Strauss 1967; Strauss and Corbin 1990), our analysis started immediately after our initial onsite visit in 2014 which guided our further data collection. For example, we decided to focus on the products and challenges that were more "strategic" for Com-Car and that required significant involvement with the community, that is, the Mud Warrior, Aqua and Terra Cars, Moja, and Wheeler (pseudonyms). During the rest of our ethnography, we continuously went back and forth between data collection, data analysis, and literature to find support or detect inconsistencies between new ideas and our data (Strauss and Corbin 1990).

More specifically, the analysis was conducted by three authors, one of whom was part of the data collection team while the other two brought fresh eyes to the data. Each researcher read through all the fieldnotes and interview transcripts. Then we selected a subset of data to open code independently. During open coding, our analysis team met once per week to compare and discuss codes until a relatively stable codebook emerged. The parent codes in this initial codebook included: the locus of design (i.e., whether the design activity was carried out jointly by Com-Car and community members or by Com-Car or community members alone), perspectives of Com-Car employees on community members, perspectives of community members on Com-Car employees, and misalignment between Com-Car and the community, among others. After further developing the initial codes, we imported them and all data into a qualitative analysis software which allowed us to independently code new data. We met regularly to discuss our interpretations of the data and finalized our codebook as our interpretations converged.

During the coding process, we wrote memos and continued regular meetings to discuss reflections on the data and potential connections to existing theory (Charmaz 2006). We used the approach of temporal bracketing suggested by Langley (1999) to segment our data into different relevant periods of time and thus develop a process theory. Periods were chosen so that there was "a certain continuity in the activities within each period and there [were] certain discontinuities at its frontiers" (Langley 1999, 707). To identify relevant temporal brackets, we took the following steps. First, during data collection we recorded changes in strategic focus or other major shifts in activities that the organization was undertaking. For example, during one of our meetings with Com-Car leadership, we became aware of a significant new project that would be introduced as a new type of challenge on the open online platform which marked an important change in the organization and its interactions with the online community. Second, we continuously monitored the online platform to detect major changes in activities and boundary relations. And, third, as we collected data, the research team met periodically (at least once every 2 weeks) to discuss similarities and differences across observed activities at any given time.

We used the analysis software to build query tables that examined the locus of design across product development periods and vehicle development programs. Co-design occurred, by our definition, in each instance coded as "joint locus of design." In doing so, we noticed that the locus of design appeared to oscillate between a joint locus (Com-Car plus the community) and Com-Car on their own. To better understand these co-design dynamics, we reviewed and open coded all quotes within the "joint locus of design." Themes emerged around different types of co-design and activities that appeared to support co-design, such as transparent exchange of design ideas, valuing community members' contributions, sharing personal design projects, etc.

We began axial coding (Strauss and Corbin 1990) by returning to the literature on co-design to help make sense of the different types of co-design and related supporting practices. This led us to Rouse's (2020) theory of "intimate co-creation" which proposes that, through supporting activities, co-creative dyads develop a "shared interpersonal boundary," that is, a "shared sense of 'we'." While it was unclear whether Com-Car employees and community members ever developed a shared sense of "we," we did see that co-design activities in some periods were stronger than in others. Inspired by Rouse (2020), we labeled the frequent, transparent, and relatively unbounded co-design activities we observed as *intimate* co-design. Then we characterized the less frequent, more strategic, and more bounded co-design activities as *instrumental* co-design. Activities that limited the community's involvement and placed Com-Car as the primary locus of design were labeled as *limited* co-design. We began to connect supporting practices to these emergent types of co-design. In doing so, we further analyzed the practices through the lens of boundary work and the three categories identified by Langley et al. (2019): collaborative, configurational, and competitive boundary work. For instance, we observed activities tied to intimate co-design like transparent exchange of design ideas and valuing community members' contributions as instances of "downplaying the boundary" within Langley et al.'s (2019) conception of collaborative boundary work.

We moved to selective coding (Strauss and Corbin 1990) to examine what was triggering the shifts toward and away from the three types of co-design and associated boundary work. We ultimately searched for cues that preceded the shifts and first categorized them into four aggregate codes. For example, we coded *shifts in strategic focus* as instances when Com-Car's employees described changes to the firm's organizational, product, or market strategy. We used the label s*hifts in technology* whenever Com-Car's employees referred to a specific technology that they might like to use in the development or production of their vehicles (e.g., 3D-printing or new materials). We labeled *shifts in the business model* when Com-Car's employees described a need to make vehicles more profitable or increase profits in other ways. We coded *shifts in community composition, size, or expectations* when changes in community makeup, competencies, or expectations were described. Finally, we coded *conflict* as any stories or complaints about community members, Com-Car employees, or community management. We further aggregated these codes into three higher level forces that we labeled environmental/market forces, strategic orientation of the firm, and co-creation community readiness.

This analysis led to a theoretical model which we developed in relation to existing perspectives on boundary work, innovation, and organizational behavior. We present our empirical model in the Findings section (Figure 2) and our theoretical model in the Discussion (Figure 4).



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## 4 | Findings

Our findings show co-design dynamics over time between Com-Car and its co-creation community. We introduce three types of co-design–*intimate, instrumental*, and *limited co-design*—and show how these coincided with the three types of boundary work identified by Langley et al. (2019). Specifically, we found that *collaborative, configurational*, and *competitive boundary work* co-existed with intimate, instrumental, and limited co-design, respectively (Table 1). We also identify environmental, community, and internal organizational forces that occurred alongside shifts between types of boundary work and associated co-design. These forces involved changes in the market and relevant technologies (i.e., environmental dynamism or profit pressure from investors), community competencies and commitment (i.e., community readiness), and Com-Car's internal strategy (i.e., moving toward a more exploration vs. exploitation-oriented innovation strategy). The following sections describe these driving forces across six periods of time and how they dynamically shifted Com-Car and its community between the three types of boundary work and associated co-design (Figure 2).

### 4.1 | Period 1: Collaborative Boundary Work and Intimate Co-Design

#### 4.1.1 | Forces Instantiating Collaborative Boundary Work and Intimate Co-Design

Period 1 begins in 2007, the year that Com-Car was founded. Com-Car was created on the promise of co-creating innovative boutique motor vehicles with an open online community. This was a strategic choice led by the Com-Car founder and CEO and part of a larger context of positive hype surrounding co-creation in the broader market at that time, a sign of environmental dynamism around co-creation. Com-Car's CEO envisioned a community of designers eager and able to contribute to co-creation who might find in Com-Car's open online community a place to share their ideas, showcase their competencies, and, perhaps, turn their designs into reality. This co-creation hype in the market, exploration-oriented innovation strategy, and growing community of enthusiasts ready to contribute their ideas came together to support an extended period of intensive design collaboration between Com-Car and their co-creation community. Between 2007 and 2014, Com-Car offered frequent open-ended design challenges and collaboratively developed designs with community members resulting in two primary vehicles, the Wheeler and Mud Warrior (Figure 1), arguably the most successful products in Com-Car's history. The online platform and in-person Com-Car fabrication area were also home to a wide range of Com-Car employees and community members' personal design projects. The intensive design collaboration between Com-Car and its community members observed during this period is what we named *intimate co-design*, a kind of co-design practice that was enabled by collaborative boundary work.

#### 4.1.2 | Collaborative Boundary Work

During Period 1, Com-Car engaged in a variety of collaborative boundary work practices that involved embodying or downplaying (Langley et al. 2019) the boundary between Com-Car and its open online community (Table 1). These collaborative boundary work practices included Com-Car valuing the community's design contributions, investing in filler activities to keep the community engaged, advocating for community members, and transparent communications between Com-Car and community members. During Period 1, Com-Car had a policy of transparently posting all emerging design ideas, decisions, and information on their open online platform so that community members could follow along and contribute. Transparency was an intentional choice intended to keep community members engaged and facilitate more opportunities for collective creativity. An observation in our field notes with Elliot, a Com-Car engineer, provides an example of transparent design exchange. Elliot told us that Com-Car was doing "a lot of experiments" with 3D-printers. A researcher asked him if experiments like these were being done elsewhere, and Elliot told us:

> Of course, a lot because everyone wants to understand how to make 3D-printing work. But they don't post their results. We post to the community everything we do. We are fully transparent. Sometimes community members contact us outside of the platform, e.g., on email or Facebook, but we always move the conversations onto the platform because we want to be transparent.

In addition to transparent communication, Com-Car hosted filler design challenges to keep their community active—even challenges that were not relevant to Com-Car's primary product development interests. They encouraged personal design projects to be shared on the platform, hosted design sketching tutorials, and engaged in other activities aimed at maintaining engagement and learning among and across community members and Com-Car employees. These coalesced into another collaborative boundary work practice in which Com-Car designers routinely expressed appreciation for community members' design contributions and attempted to honor commitments to community members. As an example, Mike, a Com-Car designer, expressed concern for Dymas, the winner of the Wheeler challenge. Com-Car had decided to halt production of Dymas's design, and Mike worried about the consequences of this decision for both Dymas and the community more broadly. He worried that it could be seen as not fulfilling Com-Car's commitment to Dymas, as described in an excerpt from our fieldnotes:

> Mike explained how he felt about having to tell Dymas the news that his project has been canceled: "*It's probably going to be one of the hardest things we're going to have to do, because, you know, a community member who is seen as a leader in the community, who was the designer of the project... I'm betting about 90% that he'll drop off the community when we make that announcement*." Mike later advocated to Com-Car leadership for Dymas to be shipped the only Wheeler prototype that was made.



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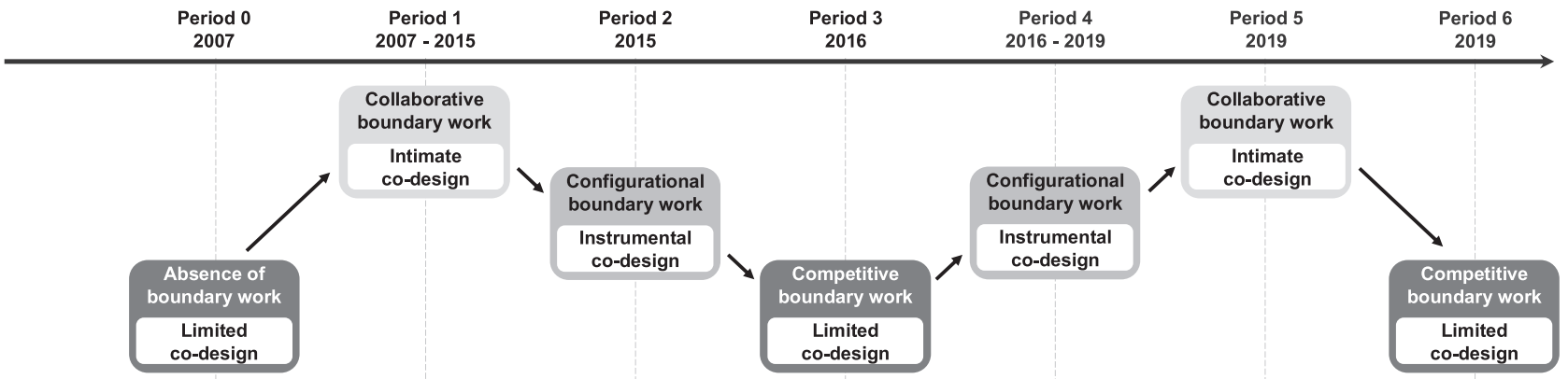


| Forces | Shifts | Absence → Collaborative boundary work | Collaborative → Configurational boundary work | Configurational → Competitive boundary work | Competitive → Configurational boundary work | Configurational → Collaborative boundary work | Collaborative → Competitive boundary work |
|---|---|---|---|---|---|---|---|
| **Environmental/ Market Forces** *(as foregrounded within Com-Car)* | *Force toward* ***Collaborative*** *boundary work* | **Environmental Dynamism–Co-Creation Hype:** Many organizations start building co-creation platforms | **Environmental Dynamism–New Technology:** 3D-printing offered potential for micro-factory-based manufacturing models | | **Environmental Dynamism–New Technology:** Rapid growth in all-electric, autonomous driving technologies | **Environmental Dynamism–New Technology:** Rapid growth in all-electric, autonomous driving technologies | |
| | *Force toward* ***Competitive*** *boundary work* | | **Unprofitable Model:** Low profits on vehicles; complex business model | **Profit Pressure:** Investors push for pathways to sales and return on investment | | | **Profit Pressure:** New investors push for sales and return on investment |
| **Com-Car's Strategic Orientation** | *Force toward* ***Collaborative*** *boundary work* | **Exploration:** Com-Car launches with a strategy focused on co-creation with a global community | | | **Exploration:** Com-Car reaffirms co-design as part of its DNA and creates a sister firm to grow & manage the community, pursuing a wide innovation agenda | **Exploration:** Com-Car reinvigorates the community by purposefully identifying and pursuing a broad set of design challenges well suited to them | |
| | *Force toward* ***Competitive*** *boundary work* | | **Exploitation:** Com-Car narrows its strategy from co-design broadly to co-design of 3D-printed vehicles & shifts responsibility for the community from their design team to marketing team | **Exploitation:** Com-Car views interactions with the community as marginal and further limits interactions; shifts focus to developing Moja 1.0; internal conflicts across Com-Car | | | **Exploitation:** Com-Car shifts from co-design of many products to development of one "real product [we] are trying to sell" [Moja 2.0] |
| **Co-Creation Community Readiness** *(community commitment & competencies)* | *Force toward* ***Collaborative*** *boundary work* | **Increasing Readiness:** *Aligned competencies and high community commitment;* community grows with many designers & car enthusiasts well suited to Com-Car's needs; community-wide enthusiasm for co-creation & Com-Car projects | | | **Increasing Readiness:** *Re-aligned competencies and renewed community interest;* the community grows again and diversifies its skills, particularly via universities well suited to meet Com-Car's evolving needs | **Increasing Readiness (Aspirational):** *Attempts to further align competencies and reinvigorate community commitment* | |
| | *Force toward* ***Competitive*** *boundary work* | | **Decreasing Readiness:** *Continued community commitment but misaligned competencies;* the community *(i.e.* designers, car enthusiasts) have limited 3D-printing expertise. Com-Car recruits engineers with 3D-printing skills to join the community | **Decreasing Readiness:** *Aligned competencies but lost commitment*; community members unhappy with Com-Car limiting interactions, changing its design challenges, and halting the production of winning designs; influential community members leave | | | **Decreasing Readiness/ Community Stagnation:** *Low community commitment.* As design challenges become sparse, active community members dwindle until the co-design platform is eventually shut down |

**FIGURE 2** | A timeline showing periods of boundary work and associated co-design between Com-Car and its co-creation community alongside the forces that appeared to drive shifts between boundary work and associated co-design types.



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**TABLE 1** | Types of boundary work and associated co-design at Com-Car.

| **Boundary work definitions** | | | |
|---|---|---|---|
| Definitions of boundary work types (Langley et al. 2019) | Collaborative boundary work<br>• Working *at* boundaries.<br>• Involves *negotiating, embodying*, or *downplaying* boundaries to enable cross-boundary collaboration (Apesoa-Varano 2013; Barrett et al. 2012; Bechky 2006).<br>• Boundaries are seen as *junctures*—a means of multiple groups joining together—and are used to facilitate mutual understanding and collaboration (Quick and Feldman 2014).<br>• Rooted in Strauss's "negotiated order" perspective (1978). | Configurational boundary work<br>• Working *through* boundaries.<br>• Involves *arranging, buffering*, or *coalescing* boundaries to manipulate cross-boundary work patterns, bringing certain activities together, while keeping others apart, to facilitate effective collective action.<br>• Boundaries are seen as both *junctures and barriers*—a means of actively managing the boundary between groups—and are used to guide the development of both collaborative and competitive interaction patterns.<br>• Described in a review by Langley et al. (2019). | Competitive boundary work<br>• Working *for* boundaries.<br>• Involves *defending, contesting*, or *creating* boundaries to elevate or ensure an occupational group's positions of authority and credibility over others (Gieryn 1983).<br>• Boundaries are seen as *barriers*—a means of defining and separating groups—and are used to acquire or reproduce a group's power, position, or resources.<br>• Rooted in Bourdieu's "fields of practice" (1977). |
| **Examples at Com-Car of boundary work and associated co-design** | | | |
| Boundary Work observed at Com-Car | Collaborative boundary work<br>• Transparent communications between Com-Car and community members.<br>• Com-Car valuing the community's design contributions and advocating for community members.<br>• Com-Car investing in filler activities to keep the community engaged. | Configurational boundary work<br>• Transactional engagement mechanisms between Com-Car and community members.<br>• Com-Car adding community managers as mediators and encouraging less frequent and more strategic communications.<br>• Com-Car bounding community members' design contributions. | Competitive boundary work<br>• Curtailing engagements between Com-Car and community members.<br>• Com-Car employees questioning the value of the community's contributions.<br>• Com-Car engineers and designers being elevated as the main sources of designs. |
| Co-design observed at Com-Car | Intimate co-design<br>• Com-Car and the community jointly designing together at the boundary.<br>• Ongoing, transparent design interactions to jointly generate and refine designs.<br>• Com-Car offering open-ended design challenges and honoring the community member's design intent. | Instrumental co-design<br>• Largely one-sided design activity, sometimes led by Com-Car and sometimes led by the community.<br>• Com-Car managing and focusing design interactions on only targeted aspects of designs.<br>• Com-Car offering tightly scoped challenges and broadly interpreting community member's design intent. | Limited co-design<br>• Largely one-sided design actively led by Com-Car.<br>• Little to no design interactions between Com-Car and community members.<br>• Com-Car largely suspending design challenges with the community. |



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These collaborative boundary work practices created an environment and relationships in which Com-Car designers and community members could engage in what we call *intimate co-design*.

#### 4.1.3 | Intimate Co-Design

We define intimate co-design as ongoing joint design work between a community and firm involving relatively open-ended design tasks, transparent design interactions, and the firm honoring community members' design intent (Table 1). Open-ended design tasks were those with a wide but intentionally crafted scope intended to facilitate collective creativity between members of the community and Com-Car. For example, Com-Car's initial 3D-printed car challenges were particularly open-ended in that they invited contributors to design any vehicle that could be both road-ready and directly digitally manufactured (i.e., 3D-printed). Once design challenges were launched, intimate co-design occurred between Com-Car and community members via the online platform, web calls, on-site interactions, and other channels, sometimes involving frequent (i.e., daily) interactions. These interactions entailed detailed feedback and suggestions shared between Com-Car employees and community members, often accompanied by artifacts such as sketches and design renderings. Com-Car employees described these interactions as helping to improve the quality of community members' design submissions.

Consider the following interaction on the online platform in which Bernhard, a community member, posts a preliminary design for the Striation challenge (see full exchange in Appendix A: Table A1). Brice, a Com-Car designer, commented: "Hi Bernhard! Nice start on the front end. I dig the fenders. What do you plan to do with the rear end?" Bernhard replied: "thanks brice! I haven't really worked out how the rear is going to look, though I'm planning a sort of open rear end area, like the Lancia 037 or Porsche 917 ... [...] I'm not sure whether I need to make the model with provisions for attaching a roof or not." Then Jack, another Com-Car designer, responded: "I would personally go without a roof and integrate the rear fenders into the body. I also have to agree with Brice that the rear part needs some support structure. From the side view it looks like it would be relatively easy to get in and out of." Bernhard shared a revised design (Figure 3, left). This sparked further ideas, including from Jeremy, Com-Car CEO, who shared: "@[bernhard]. Wow, you are pushing this in the right direction. [...] Think about steel roll-bar additions on top of the printed head-humps behind the people. Having steel right behind the heads is probably better in the case of a roll over as it deforms but doesn't totally fail in cases where the printed material might." Bernhard then shared another design rendering that incorporated many of the ideas that Com-Car employees had suggested, including Jeremy's steel roll-bars idea (Figure 3, right). We coded this design activity as intimate co-design given the ongoing joint design work and transparent communications between Bernhard—a community member—and Brice and Jeremy—Com-Car employees.

Intimate co-design also involved Com-Car honoring community members' design intent. During one observation, our field notes described Brice (Com-Car designer) working on the Aqua Car with two screens in front of him. He revised the design on his right screen while his left screen displayed the original concept designed by Rory, the community member who had won the Project Reloaded challenge. Brice explained: "I need to rework the design without changing the spirit of it. When I work, I look at it and make my changes without altering its spirit." In instances like this, Com-Car employees like Brice were quite intentional about honoring community members' design intent.

During this period, we observed strong interpersonal connections between some members of Com-Car and the community. This was evident when Com-Car employees and community members described the friendships that they had developed with one another. For example, during a field observation, Sam, a Com-Car designer based in the U.S., described how he called Dymas, a community member based in Romania, almost every day at 6 AM. Sam described how he and other Com-Car designers had sent Dymas a cage for his pet rabbit since he could not get one in Romania. Com-Car designers Sam, Brice, and Jack each shared how they had a "favorite friend" in the community whom they cared about and interacted with "a bit more." Period 1 lasted until the announcement of the Project Reloaded challenge winner which went on to become Com-Car's Terra vehicle.

### 4.2 | Period 2: Configurational Boundary Work and Instrumental Co-Design

#### 4.2.1 | Forces Moving Away From Collaborative Boundary Work and Intimate Co-Design

Starting in 2015, we observed gradual shifts in the communications and co-design that occurred between Com-Car and community members. We noticed several forces intertwined with these shifts, including increasing pressure from investors for profitable vehicles, a shift in Com-Car's strategic

**FIGURE 3** | Two renderings shared by Bernhard, a community member, showing an evolving design during the Striation challenge developed through intimate co-design between Bernhard, Brice, and Jack (Com-Car designers), and Jeremy (Com-Car CEO), among others.



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orientation from exploration toward exploitation, increased environmental dynamism with advances in large-scale 3D-printing technology, and decreasing community readiness to co-design around the emerging 3D-printing technology (Figure 2). During interviews and on-site conversations, Com-Car employees told us about the firm's low profits to date and desire to place less strategic emphasis on co-design and more on direct digitally manufactured (3D-printed) vehicles as a way to respond to the profit pressure and the environmental dynamism created by the new technology. Peter, a Com-Car designer, shared that, "the focus of the company is really not that [crowdsourcing] anymore; the focus of the company is now 3D-printed vehicles." Liam, Com-Car's CFO, explained the company's desire to still do co-design but in a more focused way that limits the "massive amounts of ideas" while still engaging "a community that gives you high-quality ideas and helps you bring them to life." These words highlight Com-Car's desire to change its strategic orientation and move the community away from an exploration-focused strategy toward an exploitation-focused one.

This change in strategy placed further pressure on the community to change and incorporate new kinds of people and expertise. As Com-Car shifted to a focus on 3D-printed vehicles, Com-Car's engineers and designers began to explore various additive manufacturing materials and processes. This required knowledge that community members at that time largely did not possess, making community members' competencies misaligned with Com-Car's needs and the community as a whole less ready to contribute to co-design with Com-Car. This led Com-Car to recruit more engineers with 3D-printing knowledge into the community and to identify existing community members with 3D-printing interest or expertise. During one of our visits, Al, a builder and engineer at Com-Car, was doing tests on additive manufacturing materials. In an excerpt from our field notes, Al explained:

> Up to now there are only a few members with specific technical skills on 3D-printing, but they are learning. They are upping up. We have different kinds of experts. Some have already done 3D-printing. Some are learning and they have good ideas. Sometimes, hidden in the community there are people who are really knowledgeable. Maybe they had not participated much before, but now they do. Some have small 3D-printers and they can do tests on small scale.

Com-Car began to expand and exploit the 3D-printing-related competencies in their co-creation community and limit efforts not aligned with this strategy. Taken together, these forces coincided with shifts from intimate to what we named instrumental co-design collaborative to configurational boundary work and from intimate to what we named *instrumental co-design*.

#### 4.2.2 | Configurational Boundary Work

During Period 2, the responsibility for engaging with community members shifted from Com-Car designers to their marketing team. The marketing group installed a new set of community management practices that strongly resemble Langley et al.'s (2019) theorization of configurational boundary work (Table 1). Com-Car structured transactional arrangements between themselves and the community, thereby manipulating cross-boundary work patterns and arranging (Langley et al. 2019) the boundary between them. In an interview, Justin, a Com-Car project manager and director of operations, illustrated this view:

> Community members are a precious resource but also a challenge for us. They have their own agenda and work, so their deadlines are not ours... To deal with that, we are thinking of putting in place a workflow transaction system. This means that we will ask for suggestions with specific deadlines. People who post their suggestions and ideas for that will receive points.

In addition to point systems, Com-Car offered financial rewards to community members based on the importance of the contributions that they made. For example, we were told that one community member was given $1500 "for one comment they made" during a design challenge.

Another way that Com-Car structured and focused the interactions between community members and Com-Car employees was the implementation of community managers as a new role on the online platform. These roles were offered to Com-Car employees and community members who had extensive experience and were influential in the community (such as Dymas, an influential community member). Community managers addressed recurrent questions and common problems faced by community members, thus avoiding the proliferation of similar requests (e.g., questions on the design brief for a challenge) and off-topic discussions (e.g., criticisms toward the idea voting system). Community managers acted as mediators and aimed to reduce the amount of what our informants called "*noise*" on the platform while keeping the conversation focused on the challenges and products most relevant to Com-Car. This is in contrast to activities like offering sketching tutorials or filler design challenges that aimed to promote a joint community of practice but did not directly address Com-Car's product needs, as described in the practices of collaborative boundary work.

Also, during this period, Com-Car employees began communicating strategically with community members on a need-to-know basis. Com-Car enacted policies that limited communications between Com-Car and community members and discouraged posting on the open online platform except for information that the community needed to know. Consider the example of Justin, a Com-Car engineer and manager. He described needing to design brackets to support neck rests for the Aqua Car, and when we asked about the community's involvement in helping design the brackets, Justin explained that the community was not involved in the Aqua Car. He shared that since Com-Car didn't yet know how to build a fully 3D-printed road-ready car, they used engine parts from an existing vehicle to make it work in time for their target car exposition. To keep this information private, Com-Car designers were instructed to limit communications with community



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members during the Aqua Car design process. In addition to strategically obscuring technical information, limiting the exchange of design ideas was also used to direct community members to focus on what Com-Car deemed as "critical path" design tasks that best served Com-Car's needs.

#### 4.2.3 | Instrumental Co-Design

Configurational boundary work practices supported what we define as *instrumental co-design*—instances when Com-Car employees offered community members tightly scoped design tasks, broadly interpreted community members' design intent, and engaged in a more limited and focused exchange of design ideas with community members (e.g., once per design challenge). In contrast to the open-ended design tasks of intimate co-design, Com-Car invited community members to engage in narrowly defined design tasks and tightly scoped design challenges, such as collecting information about a specific material or providing feedback on a vehicle concept conceived by Com-Car employees. After completing the Mud Warrior concept design, for example, Com-Car launched several small challenges aimed at refining specific parts of the car and helping Com-Car to launch the vehicle. In 2015, Com-Car launched the Mud Warrior Artwork challenge aimed at creating multiple new "looks" for the car. The Challenge Brief stated: "Now that the Mud Warrior is almost complete, it is time to pay homage to all the shortened pencils and emptied pens, by creating a collection of great Mud Warrior artwork. Your challenge: Create a beautiful and exciting rendering of the Mud Warrior and its universe. This image should celebrate the heritage, possibility, and purpose of our first vehicle." This relatively narrow design challenge resulted in multiple "skins" for the exterior of the Mud Warrior vehicle that Com-Car went on to commercialize. As Com-Car engaged its community in more tightly scoped design tasks, the frequency of Com-Car-community design interactions became intermittent. During this period, Com-Car and community members tended to interact only once or twice around a design instead of frequently over a sustained time as was the case in intimate co-design. Additionally, rather than valuing and honoring community members' design intent as Com-Car employees did during Period 1, there were times during Period 2 when Com-Car encouraged its employees to re-interpret community members' design intent. For example, in our field notes we observed that after the Aqua Car concept was selected for further development, Craig, Com-Car's head of product development, held a staff meeting and asked Com-Car designers and engineers to generate a new structure for the vehicle, one that could be manufactured, and to specifically not consider the community member's design as an "*anchor*."

### 4.3 | Period 3: Competitive Boundary Work and Limited Co-Design

#### 4.3.1 | Forces Moving Away From Configurational Boundary Work and Instrumental Co-Design

In 2015, Com-Car entered a period characterized by new forces, including additional pressure from investors to sharpen Com-Car's strategic focus on profitability and community forces involving conflict within and between Com-Car and the community and influential community members leaving the community (Figure 2). Community members had been expecting to see their designs come to life, at least in the form of a prototype, regardless of manufacturing costs. After three vehicles with limited profitability (Wheeler, Mud Warrior, and Terra), Com-Car's CEO began to stress the need to profit from vehicles and outgrow their "*constant prototyping*" phase. Community members complained about changes in the way Com-Car engaged with them, which came to a head in 2016 during Project Reloaded. This was the first design challenge launched by Com-Car's marketing team instead of Com-Car's design team. The content and scope of the challenge did not resonate well with community members, which exacerbated frustration across the community, and conflict within Com-Car and between Com-Car and the community. For example, some community members expressed disappointment, frustration, and even anger about the objectives and content of the Project Reloaded challenge. After a long exchange of "*flaming*" posts, one community member wrote:

> Looking at the comments here [...], I found myself relieved about my own complaints and concerns on this project/challenge ... there's a lack of information in all sense that's worrying. Indeed, where's the case study for a car like this? who is the target consumer? where are the powertrain specs for design consideration? where are the marketing foundations for a project like this? ... well, for sure at the end I will finish with more questions rather than answers ... and as other colleagues said before, I'm not sure to keep the push on my own design development or better quit before a major frustration or anger ...

This community member was not alone in their sentiments. The percentage of conflictual interactions (the ratio of posts with disagreements and open conflicts to the total number of posts in a challenge) between Com-Car employees and community members during Project Reloaded was about 8 times greater than the conflictual interactions during Striation, the previous challenge. Such a conflictual climate further diminished participation and commitment to Com-Car's hosted challenges, thereby decreasing the community's readiness to contribute to design challenges. Conflict with and within the community and increased profit pressure came together to push Com-Car toward an exploitation-oriented strategy. This coincided with a shift from configurational to competitive boundary work and from instrumental to what we call *limited co-design*.

#### 4.3.2 | Competitive Boundary Work

During Period 3, we observed several competitive boundary work practices that created and defended (Langley et al. 2019) boundaries between Com-Car and the community (Table 1). Com-Car employees questioned the value of community members' contributions. Liam, for example, described in an



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interview that no one at Com-Car really knew what co-creation bought them and that it was questionable whether each dollar they put into co-creation really generated a favorable financial return. He asked: "Maybe it would be better to do all our innovation in house?" Even Peter, a long-time proponent of co-design within Com-Car, shared his opinion that, out of all submissions from the community for Project Reloaded, they had received "only five decent designs." In addition to questioning the value of the community, Com-Car implemented interaction protocols through the marketing and legal teams that, as one designer described, "shut off" communication between Com-Car designers and community members. During one of our observations, Jack, a Com-Car designer, explained how he was no longer doing any project updates in the community. He told us that all he wrote were "*superficial comments like 'cool, dude'*." Ever since he had overheard how Craig was yelled at by Lindsey, a marketing team member, he didn't dare write anything on the community anymore because he didn't want to find himself in a bad position. The problem, according to Jack, was that the communication with the community had been shut off and not replaced with anything. As a result, many leading community members had left. Taken together, these competitive boundary work practices created and fortified a boundary between Com-Car and the community and elevated Com-Car employees as the main source of Com-Car designs.

#### 4.3.3 | Limited Co-Design

We define *limited co-design* as occasions when the co-design platform remained open, but employees and community members rarely or never used it. In these instances, Com-Car provided few or no design tasks to, and interacted rarely with, or actively excluded, community members. Consider the conversation we observed between Peter and Brice, two Com-Car employees. They were discussing the structure of the community and a challenge that was to be launched and lamented: "Now we are asking questions to a community we don't have." They asked rhetorically: "What are you recruiting them to?" and reflected that the community had "*deteriorated*." There were times during this period when Com-Car offered no design challenges to the community, which led to little or no engagement between Com-Car and community members. The number of comments that we observed on the open online platform decreased by about 70% during Period 3. Com-Car designers were no longer allowed to communicate directly with community members, and any messages from Com-Car to the community had to be approved and posted from the Com-Car marketing department, which led to much less frequent communication. However, this period of competitive boundary work and limited co-design at Com-Car did not last long.

### 4.4 | Period 4: Configurational Boundary Work and Instrumental Co-Design

#### 4.4.1 | Forces Moving Away From Competitive Boundary Work and Limited Co-Design

While Period 3 involved severely curtailed communication and co-design between Com-Car and the community, Period 4 involved renewed attempts to bring the community back to the forefront of innovation at Com-Car alongside a more exploration-oriented innovation strategy (Figure 2). In late 2016 and early 2017, we observed new attempts to solicit community members' commitment and design ideas through focused design tasks. This coincided with renewed environmental dynamism represented by new technologies and market growth in all-electric and autonomous driving, representing a force on Com-Car to develop new innovative solutions for mobility. At this time, Com-Car also made the strategic choice to re-affirm co-design as a crucial part of its "DNA" and explore new vehicle opportunities through the open online community. This strategy was, in part, driven by Com-Car's need for new competencies around new forms of public transportation, autonomous mobility, and continued advances in 3D-printing. In our field notes, Curtis, a Com-Car engineer in charge of university collaborations and hackathons, said that despite collaborations sometimes being challenging, finding the right partners was essential to advance Com-Car's 3D-printing competencies. Jack, a Com-Car designer, acknowledged a need to "diversify the targeting effort" as Com-Car recruited new community members to meet its ever-evolving needs. Once again, Com-Car tried to reconstruct alignment between the community's competencies and the company's needs in an attempt to increase the readiness and willingness of community members to participate. To better manage these issues, Com-Car decided to spin out a new firm to manage their open online community which formalized their move back to configurational boundary work.

#### 4.4.2 | Configurational Boundary Work

During a field observation, we were told by community managers at the sister organization operating the open online platform that they had begun to strategically "monetize" the community as a service. The platform was renamed and separated from Com-Car in ways that signaled that the community would now receive challenges from a wider set of organizations, not just Com-Car. Communication between community members and Com-Car (including the new spin-out firm) was limited to strategic, need-to-know information with relatively transactional interactions. We heard from multiple Com-Car employees in our field notes that they viewed the community as a service and regularly requested tightly scoped design contributions or user inputs focused on only Com-Car's prioritized needs. This marked a transition back to configurational boundary work and instrumental co-design (Table 1).

#### 4.4.3 | Instrumental Co-Design

Com-Car engineers and designers began giving the revived community tightly scoped design tasks for the vehicles that they were developing. For example, while Com-Car was designing the front end of Moja, an electric autonomous bus, multiple design challenges were posted online to define the "eyes" of the bus. During a field visit, Jack, one of Com-Car's designers working on the project, explained to us:

> In order to collect data from users, we now collect data [from community members] through interviews and surveys. For instance we ask: which is the gender you would prefer to assign to Moja? The marketing team



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> was thinking of giving Moja a feminine gender in line with the female voices of Alexa, Google, and the first GPSs. However, our analyses [with community members] show a preference for a male voice or for no gender at all.

In addition to surveying community members about their preferences as potential users—a new way that the community was "used" instrumentally—Com-Car posted focused design challenges related to Moja's eyes. Jack further explained how community members designed different versions of "the 'robot eyes' in the front to look 'cute'. They ended up with cute eyes, but not feminine eyes." The community was also given design challenges in collaboration with other companies, like delivery drones in adverse conditions or mobility on Mars. Our observations on-site and analyses of the online posts revealed that, during Period 4, interactions between employees and community members were somewhat infrequent and extremely focused. Com-Car employees typically specified their problem or design challenge but did not engage in repeated interactions with the same community members over time. Additionally, in one of our visits in 2018, we learned that only three out of over one hundred Com-Car employees were still members of the platform. Everyone else had left or was not expected to join and interact with community members.

## 4.5 | Period 5: Collaborative Boundary Work and Intimate Co-Design

### 4.5.1 | Forces Moving Away From Configurational Boundary Work and Instrumental Co-Design

In 2019, in the wake of the environmental dynamism from interest in all-electric vehicles and autonomous driving, Com-Car made the strategic choice to lean further into an exploration-oriented strategy and further reinvigorated its community and co-design interactions. Com-Car staff told us how their monetized, instrumental co-design approach had become limiting, and how they aspired to further strengthen the commitment and contributions of community members, including a growing set of university students. George, one of the community managers, described in our field notes: "We are not Mechanical Turk or another crowdsourcing platform. [...] We need to get closer to our community members ..." This marked a shift in internal strategy back toward engaging in collaborative boundary work and intimate co-design (Figure 2).

### 4.5.2 | Collaborative Boundary Work

Com-Car attempted to strengthen collaboration at the boundary between the firm and community by investing again in collaborative boundary work (Table 1). They began offering design challenges for the sole purpose of reinvigorating the community. For example, during Period 5, Com-Car offered a filler challenge related to aerial drone swarms. Jack, a Com-Car designer, described how Com-Car "has no plans to work on drones" and did the competition just to keep the community active. He also described "pop-up lots," another filler design challenge that explored uses for parking lots in a future when cars are no longer needed. Again, this challenge was offered "just to keep the community active." Com-Car also began to explore engaging community members in problem framing, not just solution generation, to give the community more voice in the design process. George, a community manager in the spin-off firm, shared in an interview that "in the past we used to go out to clients and develop a design brief for each challenge. Now we want to involve the community in framing the problems."

### 4.5.3 | Intimate Co-Design

This renewed collaborative boundary work helped to spark renewed intimate co-design activities through the online platform and collaborations with universities. New initiatives were implemented on the online platform, such as discussion forums where community members and Com-Car employees were invited to organically share and discuss new ideas. Com-Car experimented with offering different types of tasks and challenges to the community, including challenges around popular interests among community members, such as sustainable mobility and autonomous driving. Com-Car designers and engineers also worked closely with community members on the Moja vehicle, and new partnerships were formed with faculty and students at two US universities. However, this new period of collaborative boundary work and intimate co-design was mostly aspirational and relatively short-lived.

## 4.6 | Period 6: Competitive Boundary Work and Limited Co-Design

### 4.6.1 | Forces Moving Away From Collaborative Boundary Work and Intimate Co-Design

Later that year, Com-Car's CEO raised a new round of funding that was contingent on developing and selling Moja 2.0, Com-Car's latest autonomous shuttle. This represented a renewed pressure for profits that created strong forces to demonstrate a "real product [we] are trying to sell," as one Com-Car employee described. As community readiness for co-creation in the previous period was largely aspirational, the community stagnated and did not show strong commitment or participation in Com-Car's challenges. In Period 6, Com-Car took a hard turn in its strategic orientation toward exploitation and focused on exploiting ideas and design capabilities within the company. This preceded a shift away from community engagement and a dwindling of the open online community (Figure 2).

### 4.6.2 | Competitive Boundary Work and Limited Co-Design

During this period, in our observations, Jack, a Com-Car designer, explained that Com-Car employees were so focused on developing Moja 2.0 that they started outsourcing their engagements with the community. They actually hired an external engineering design firm to complete a challenge with community members. Com-Car employees' views of the community also shifted. During one of our final site visits, we asked Peter and Patrick, two Com-Car engineers, about the role of the community. Peter told us that he was skeptical about the role of the community. He was unsure



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community members could really provide useful designs for the level of technological complexity of Moja 2.0. Patrick described an example design challenge that had recently been given to the community and concluded that, "the result was obvious and not new at all. Anybody could have designed it." Patrick questioned the value of the community, adding that most community members were "below average" in terms of expected contributions and competences. These beliefs signaled a shift back to competitive boundary work—where boundaries were created and acted as barriers between Com-Car engineers and community members, thus elevating the positions of Com-Car engineers and devaluing the contributions of community members (Table 1). Not long after these conversations, Com-Car's in-person collaborations with universities and other partners were significantly reduced, and Com-Car's open online community platform and associated spin-off were permanently shut down. This marked the end of an era of co-design at Com-Car.

## 5 | Discussion

### 5.1 | Toward a Theory of Co-Design Dynamics Between Companies and Communities

The present study offers new insights into the evolution of co-design efforts at the boundary between a firm and its open online community. It does so by introducing multiple forms of co-design–intimate, instrumental, and limited–that describe the tasks and interactions jointly carried out at the boundary between the firm and its online community. It further shows how the firm evolved in how it engaged in boundary work and management of community interactions to drive and sustain co-design over time, speaking to three previously identified types of boundary work (Langley et al. 2019)–collaborative, configurational and competitive—and how these are intertwined with our three introduced forms of co-design—intimate, instrumental and limited, respectively. Ultimately, although we cannot claim causal ties from our data, our grounded theory identifies a set of intertwined external and internal forces that appear to coincide with the shifts between forms of co-design and associated boundary work. These factors, both internal and external, include changes in the market environment, community readiness, and strategic orientation toward innovation. In some scenarios, when the environment became more dynamic, the community was "ready" in terms of competencies and commitment, and the organization's strategy was geared toward exploration, we observed shifts from more competitive toward more collaborative boundary work and associated co-design processes. In other scenarios, when the environment placed greater pressure on profits, the community was not "ready" in terms of competencies or commitment, and the organization's strategy was geared more toward exploitation, we observed shifts from more collaborative toward more competitive boundary work and associated co-design processes. Figure 4 provides a formal theory that abstracts from our grounded model in Figure 2 to represent these shifts and their driving forces.

### 5.2 | Theoretical Contributions

The present study offers three main theoretical contributions: (1) illuminating the dynamics of co-design, including how and why a firm's engagement with an open online community shifts between different forms over time, (2) expanding design theory to suggest intimate and instrumental co-design as subcategories of co-design within existing co-constructive and participatory design approaches, and (3) illuminating the temporal dynamics of boundary work and contextual forces that drive shifts between boundary work types, including environmental dynamism and profit pressure, exploration versus exploitation strategic orientation, and co-creation community readiness, in effect theorizing the arrows in Langley et al.'s integrative model of boundary work (2019).

#### 5.2.1 | Explaining the Shifts in Co-Design and Boundary Work Between a Firm and a Community

First, we draw attention to the *dynamics* of co-design and identify forces that appear to instigate different forms of co-design—dynamics that have been suggested for future study (e.g., Jeppesen and Lakhani 2010; Faems et al. 2010; Stanko et al. 2017; Belderbos et al. 2010; Laursen and Salter 2006; Gemser and Barczak 2020). Scholars in the design field suggest that co-design may oscillate back and forth between "strong and governing" and "weak, consultative, and extractive" kinds of community participation across time and design phases (Costanza-Chock 2020, 91; Elsayed-Ali et al. 2026; Luck 2018). However, the scant empirical evidence on how co-design efforts in open innovation evolve over time, not only to generate value through more ideas but also to capture value (e.g., West and Bogers 2014), leaves us with a limited understanding of the dynamics of co-design. Our study begins to fill this gap by showing both *how* and *why* one firm's engagement with a community shifted between different types of community engagement and co-design activities over time.

Regarding the *how*, our theoretical model shows how firms and communities dynamically move between three types of co-design via changes in their boundary work practices. Our micro-level focus on collaboration at the boundary and co-design activities between organization and community members enables us to grapple with the micro-foundations of various community management practices and levels of community participation. This approach advances knowledge about the evolution of open innovation practices (Pinarello et al. 2022). Additionally, it helps scholars to overcome the dichotomous focus in prior studies that often examine either the online community or a firm's community management practices in isolation (e.g., Liu et al. 2022; Lindberg et al. 2022; Pinarello et al. 2022). When moving toward a more competitive boundary, firms tend to centralize decision-making and streamline interactions with external partners. While this approach facilitates clarity and control, it can inadvertently constrain co-design efforts and diminish active collaboration with a community. Conversely, when moving toward a more collaborative boundary, organizations decentralize and foster external relationships. This approach reinvigorates collaboration and renews intimate or instrumental co-design efforts but requires investment in collaborative or configurational boundary work. These findings contribute to the growing understanding of the interplay between community management practices and organizational evolution in open innovation contexts (Randhawa et al. 2017; Mahr et al. 2014).



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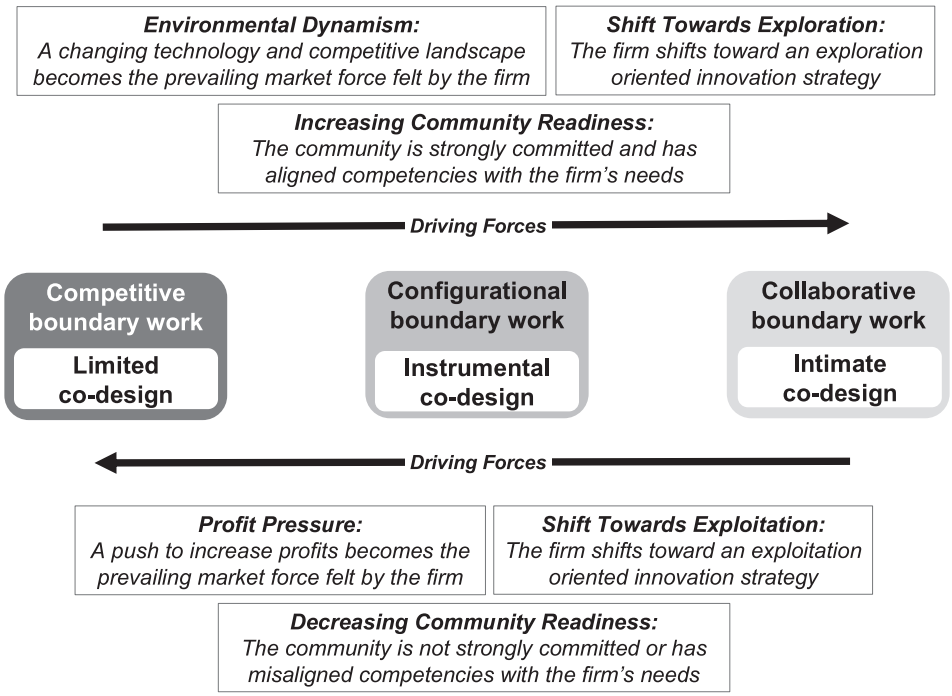


**FIGURE 4** | A dynamic model of co-design, including the environmental and internal forces driving shifts between types of boundary work and associated co-design.

Our study further offers an explanation for why co-design evolves over time by showing a set of forces that appear to coincide with the shifts between different forms of co-design and associated boundary work, thus complementing prior discussions on the factors affecting the evolution of open innovation practices over time (Pinarello et al. 2022). In line with the broader innovation literature that distinguishes between external and internal forces shaping an organization's innovation efforts (e.g., Popa et al. 2017; Chang et al. 2011; Prajogo and McDermott 2014), our findings highlight two key external forces, that is, changes in the market and technological environment and changes in community readiness, as well as one internal force: shifts in strategic focus between exploration and exploitation (see Figure 4).

Among the two external forces highlighted by our data, particularly salient is the emergent concept of community readiness, which reflects both the commitment and competence of community members as perceived by the firm. A reduction in community readiness appears to lead organizations toward more competitive boundaries and limited co-design with external stakeholders. Conversely, when community readiness increases, there is a notable shift toward more collaborative boundaries and deeper co-design practices, characterized by mutual trust and shared decision-making. This notion of readiness aligns with leadership contingency theories, particularly Situational Leadership Theory (SLT) (Blanchard et al. 1993; Thompson and Vecchio 2009). According to SLT, when a group demonstrates high readiness, defined by both high competence and commitment, a delegating leadership style is most effective, allowing members greater autonomy (e.g., Thompson and Glasø 2018). In contrast, when readiness is lower, leaders must adopt more directive or supportive styles, focusing either on task structure, relationship building, or both (e.g., Zigarmi and Roberts 2017). While the focus of this paper is not on leadership practices per se, the managerial practices we observed in navigating community dynamics closely mirror the logic of SLT. The firm adapted the engagement practices in ways consistent with the group's perceived readiness, that is, providing more structure and guidance when commitment or competence was lacking, and stepping back to enable co-design when readiness was high. It is important to highlight that it was the perceived, rather than the actual, community readiness that served as the primary driver of the shift in boundary work. A striking example of this is the shift observed between Periods 4 and 5 in our data. During this time, the company perceived the external community as being both committed and aligned with the new skill sets required by the organization. However, this perception turned out to be more aspirational than evidence based. Unsurprisingly, the shift toward deeper collaboration proved short-lived, as the anticipated readiness of the community did not materialize in practice. Future research might further explore how readiness shapes the appropriate balance between control and collaboration in boundary-spanning innovation efforts.

An internal force associated with shifts in boundary work was the innovation strategy elected by the company. We observed that shifts toward more exploitation were related to more competitive boundaries and less co-design, while shifts toward exploration were tied to more collaborative boundaries and intimate co-design. Research supports a complementary relationship between open innovation and organizational ambidexterity, that is, an organization's ability to sustain both exploration and exploitation (e.g., Srisathan et al. 2023; Michelfelder and Kratzer



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2013). For example, van Lieshout et al. (2021) and Yana et al. (2022) find that combining ambidextrous strategies with open innovation attributes is associated with competitive advantage and sustainability.

However, previous work tends to adopt a "static" view of ambidexterity (Cao et al. 2024) and does not often consider how shifts in an ambidexterity "mode," toward either exploration or exploitation, may influence changes in open innovation practices. Our theory extends this perspective by suggesting that an increased emphasis on either exploitation or exploration, rather than maintaining a balance between the two, can be a key driver of changes in the boundary work between a firm and its community. A promising avenue for future research lies at the dynamic intersection between ambidexterity and boundary work in open innovation.

It is also important to underscore that the three main forces identified in our theory are multilevel and interdependent. Environmental-level forces are closely tied to internal strategic decisions, particularly around innovation strategy. In turn, these innovation strategies may both influence and be influenced by the readiness of the community, creating a recursive loop that shapes how boundaries are constructed and maintained.

#### 5.2.2 | Types of Co-Design Between a Firm and a Community

As a second theoretical contribution, our study proposed types of co-design that build upon existing theory in the design literatures and suggest subcategories of co-design within co-constructive (Mazzurco et al. 2018) and participatory design (Kensing and Blomberg 1998; Sanders and Stappers 2008; Wacnik et al. 2025). Specifically, our findings detail two main forms of co-design between a firm and an open online community—intimate and instrumental co-design. While studies in the design literatures well describe participatory design approaches and how these vary across different levels of stakeholder engagement and decision-making power (e.g., Arnstein 1969; Mazzurco et al. 2018; Sanders and Stappers 2008), our study shows how co-design activities can vary in their relational versus transactional nature, including the level of design task openness, transparency of design interactions, frequency of interactions, degree of honoring community members' design intent, etc. In doing so, we contribute to a better understanding of co-design, in general, and in the context of online communities, in particular.

Moreover, our concept of "intimate co-design" builds upon Rouse's (2020) theory on "intimate co-creation" between creative dyads (think John Lennon and Paul McCartney or Marie and Irène Curie). Rouse theorized that creative pairs engage in practices, perhaps akin to collaborative boundary work, that develop and maintain a "a shared sense of 'we'" that supports "unfiltered idea disclosure, supportive idea elaboration, and idea evaluation that is mutually focused on the idea" (2020, 182). Our study broadens the concept of intimate co-creation beyond the dyadic level, showing how it applies to co-design between an organization and an open online community.

#### 5.2.3 | Understanding the Temporal Dynamics of Boundary Work

Our third theoretical contribution is illustrating the temporal dynamics of boundary work and connecting design and innovation management literatures to boundary work literature which, to the best of our knowledge, has not previously been done. In doing so, our results contribute to studies on boundary work. In their review, Langley et al. (2019) call out temporal dynamics of boundary work and boundary work in new organizational forms as avenues for future research. They claim that "more fine-grained work is needed to authentically capture the sayings and doings that people engage in to influence demarcations shaping their social context" (Langley et al. 2019, 56). In response to this and other calls in the innovation management literature (Pinarello et al. 2022), our findings illustrate the micro-level happenings of collaborative, configurational, and competitive boundary work in an open innovation context. In our case, these different forms of boundary work are also associated with different types of co-design. This, by itself, is an important finding: co-design represents the design interactions at the boundary between a firm and community, and boundary work consists of the boundary negotiating, embodying, downplaying, arranging, creating, and defending practices to sustain different forms of co-design. Our empirical and theoretical models describe not only how co-design, but also boundary work practices, shift over time.

Going further, we identify contextual forces associated with shifts between boundary work types, in effect, theorizing the arrows in Langley et al.'s integrative model of boundary work (2019). Previous studies on boundary work have acknowledged the importance of local conditions and contextual factors in shaping boundary work (Liberati 2017; Meier 2015). Our study adds to this discussion by identifying a set of forces shaping boundary work and showing how shifts toward competitive boundary work might be due to organizational attempts to move toward an exploitation-oriented innovation strategy while addressing environmental dynamism, profit pressures, and limited community readiness. This contribution offers a link between boundary work and innovation studies.

Lastly, our study responds to Langley and colleagues' call by revealing practices of boundary work as they occurred in a non-traditional organizational arrangement—that of open innovation with an online community. While most studies of boundary work examine collaborations between different occupational groups (e.g., Lefsrud and Meyer 2012; Persson 2010; Bechky 2006; Howard-Grenville et al. 2017; Liberati 2017; Meier 2015; DiBenigno and Kellogg 2014), boundary work has rarely been applied to study the boundaries between a firm and an open online community (see O'Mahony and Bechky 2008, for a notable exception).

### 5.3 | Practical Implications

The present study offers implications for organizations and practitioners interested in open innovation and co-design. First, our study provides guidance on how to sustain co-design over time. Though effective boundary work can take multiple



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forms—whether collaborative, configurational, or competitive—our findings suggest that competitive boundary work can discourage co-design and limit genuine community participation. Innovation practitioners interested in intimate or instrumental co-design may then adopt community management practices associated with collaborative or configurational boundary work—for instance, communicating transparently with community members, investing in challenges and activities related to community members' interests, incorporating mediators to reduce conflict, cultivating personal relationships with community members, etc. Similarly, organizations should take care in deciding who is responsible for community engagement. Our findings suggest that, during shifts from collaborative boundary work to configurational boundary work, limiting the voice of community managers and shifting who within an organization is responsible for community engagement can be detrimental and fuel conflict at the boundary.

Our study also has implications for how to engage open communities for explorative or exploitative innovation strategies. To help open communities to be effective in exploring new ideas, organizations might propose open-ended challenges and provide regular, informal, and personalized feedback on the ideas generated throughout the co-design process. This can encourage a diversity of contributions from the community to flourish. Alternatively, if an organization takes a more exploitation-oriented innovation strategy, they might propose more targeted and focused challenges as well as more formal, structured feedback and communication throughout the co-design process.

It is important for innovation practitioners to monitor the evolution of their community over time, not only in terms of quantity or quality of contributions, but also in terms of the characteristics, needs, and expectations of community members. Such changes in composition and competencies can shift the readiness of the community to engage in co-design, particularly in relation to emerging technologies that might be considered by the organization. There can also be a misalignment in the expectations of community members and the needs of the organization. Building on community members' past experiences with the open community, they can develop preferences on specific types of innovation activities, and organizations should consider community members' voices to sustain their participation over time. In order to monitor the evolution of communities and be open to community members' voices, organizations might consider periodic surveys to elicit community members' expectations, understand community members' preferred management style, or simply track the evolving backgrounds and characteristics of their community. Organizations interested in sustaining intimate co-design should remember the importance of cultivating personal relationships with community members. This includes practices, such as those pursued by Com-Car, when it hired reputed members of the community to work not only as co-designers but also community liaisons.

Finally, maintaining co-design can involve the design and management of complex socio-technical structures on online platforms. Our findings suggest that co-design platforms may benefit from specific features that support collaborative and configurational boundary work and associated intimate and instrumental co-design practices. These might include tools for offering tutorials and skills building sessions, customizable channels for communication and visualization of content, and features that support the creation of liaison roles to mitigate conflict.

## 5.4 | Limitations and Future Research

As with any study, ours has limitations. First, our data comes from a single organization and open innovation platform. Com-Car is an extreme case or "talking pig" (Siggelkow 2007) that helped us to uncover new theoretical understanding of co-design processes. However, it does not allow us to establish causality or generalize to other contexts or organizations. That said, we suspect that the same or similar types of co-design, boundary work, and contextual forces may apply in other empirical settings, especially with knowledge intensive firms engaging in collaboration with open online communities to develop new products. For instance, companies in the fashion or design industries may similarly engage with their customers to develop new products and tailor their participation to their needed competences and willingness to craft entirely new products or refine existing ones. Future studies may also analyze different platforms and organizations, for example platforms developed by large companies or communities whose members are driven mainly by extrinsic motivators (e.g., monetary rewards and career development). Particularly interesting could be the investigation of open innovation involving the co-location of clients and stakeholders. Organizations using a design thinking approach, for example, sometimes create physical spaces for long term collaboration with clients, suppliers, and other stakeholders on their organizational premises to develop and experiment with new products (e.g., Montanari et al. 2020). Understanding how different types of co-design and boundary work unfold through face-to-face interactions, physical artifacts, and a physical space are potential areas for future research. More broadly, our findings could be relevant to understanding the co-design and boundary work between organizations and external consultants hired to spur innovation or even across different units within an organization contributing to new product development.

Our data show different forms of co-design and boundary work, though we were not able to explain if and how different forms of co-design were related to the success or failure of the firm as a whole. We suspect that one form of boundary work is not intrinsically better than another and that the right approach to nurturing a co-creative effort is dependent on the situation. Future studies may investigate associations or causal connections between the types of boundary work and co-design employed, or the sequence of these types, and the success of the co-design or open innovation effort. This could be done by comparing many projects within a platform that varies in its approach to boundary work or the experiences of multiple different co-design platforms. We would expect that, if an organization maintains its strategy, direction, and other contextual forces as we described, then shifts between different forms of boundary work may be unproductive. On the other hand, as firms experience shifting strategies and environmental forces, oscillations between different kinds of boundary work could improve co-design performance. Ultimately, better understanding such boundary work



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and co-design dynamics may help organizations and their external communities to more fruitfully design together.


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### Ethics Statement

The authors have read and agreed to the Committee on Publication Ethics (COPE) international standards for authors.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

Research data are not shared. The data that have been used are confidential.

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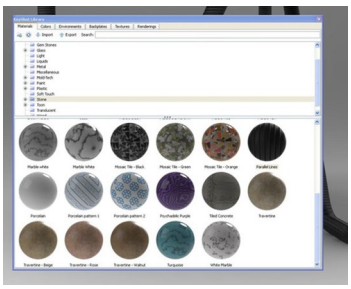
## Appendix A

**TABLE A1** | A sample online interaction illustrating intimate co-design between Com-Car employees and community members.

| | |
|---|---|
| **Brice** (Com-Car designer) | Hi Bernhard! Nice start on the front end. I dig the fenders. What do you plan t do with the rear end? |
| **Bernhard** (creator & community member) | Thanks brice! I haven't really worked out how the rear is going to look, though im planning a sort of open rear end area, like the lancia 037 or porsche 917.... however, I might cover it, since it is an electric vehicle.... we don't want any problems with rain right? :-P im not sure whether I need to make the model with provisions for attaching a roof or not. is this a compulsory requirement? I feel it would be more accessible and perhaps better looking without the roof and large windshield assembly :) |
| **Brice** (Com-Car designer) | I absolutely love the layered texture you used. It makes it very easy to understand the limits of overhang and the need for support or geometry modification when the design goes beyond these limits. Cheers! |
| **Jack** (Com-Car designer) | I would personally go without a roof and integrate the rear fenders into the body. I also have to agree with Brice that the rear part needs some support structure. From the side view it looks like it would be relatively easy to get in and out of. |
| **Jeremy** (Com-Car executive) | BOOM, the first accurate and thoughtful entry I have seen focused on the limits and advantages of the technology! There may be some unmakeable sections but nothing that could not be tweaked into shape. Keep evolving....well done Bernhard. Really well done. |
| **Bernhard** [to LM employees] | i've added an area for windscreen, and a blank area for the dials at the centre of the dash... I will soon do some provisions for the front lights, as well as the battery and motor accommodation.... how does it work tho? do we need to make the batteries accessible for removal/ replacement, or is it ok to cover the motor and battery up? |
| **Jack** (Com-Car designer) | battery and motor should be able to be installed/ serviced, but it does not have to be a swappable battery. You should go to the discussion tab at the top of the challenge brief to get updated on the limits of the tech |
| **Bernhard** | Thanks jack! :) much appreciated! |
| **Fedde** (community member) | Hi Bernhard how did you get that contour pattern done? is it multiple cut extrude sections across the whole body? really helps understand the form in the 3d print context. |
| **Bernhard** [to Fedde] | hey my guru! long time :D i've been using a material in keyshot called parallel lines.... I place the material, then use the mapping tool to change the orientation by 90 degrees into a horizontal format, then change the scale of the texture to as low as I can, so it looks like a 3D printed object :) in my laptop it comes under the plastic category, but in my office it seems to fall under 'stone' category... just look around for 'parallel lines' ..... it's available on keyshot 4 and possibly newer versions. cheers!  |
| **Jeremy** (Com-Car executive) | @[bernhard]. Wow, you are pushing this in the right direction. The pattern (fedde is right) is pushing people to see what is happening in the vehicle....SO GOOD. Here are some tips (just some thoughts). (1) I think that the nose can use some aesthetic shaping, It does not need the opening in the front, I think, and you can feel free to explore some new shapes. (2) Play with some surface areas that have been milled to texture either for pattern or for fitment of tight tolerance components....for example, you might have a milled border around the light inserts in the front so that it is clear that "holes" such as the place in which you insert components like that are prepared by a milling machine. Same for dash components like a cell phone. Further, you can use the milling capability to provide other surface treatments where necessary or pleasing to look at. (3) Think about steel roll-bar additions on top of the printed head-humps behind the people. Having steel right behind the heads is probably better in the case of a roll over as it deforms but doesn't totally fail in cases where the printed material might. (4) I love the see-through idea to show the inner walls or stantions.... you might connect them with certain cross members that are triangulated to provide side-impact protection. |

(Continues)



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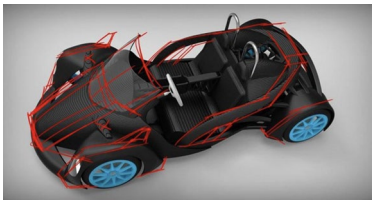
**TABLE A1** | (Continued)

| | |
|---|---|
| **Bernhard** | thanks for the valuable input jeremy! :D I will try out whatever I can as long as my laptop can take it :-p |
| **Brice** (Com-Car designer) | The best option we are considering at this point is assembling and disassembling batteries from underneath the car. |
| **Brice** (Com-Car designer) | The elevated view is gorgeous. |
| **Bernhard** | So guys what do you think? .... rounded snout or the older version stubby front with an opening that goes all the way to the rear?  |
| **Fedde** (community member) | excellent work, Bernhard, excellent. I think this is a very personable and practical interpretation of the brief. some very well grounded thought gone into this. may I suggest some toning to some shapes? |



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